CLOUD EDUCATION LEARNING (P) LTD
Registered Address: K No- 2121041180X, Chakkarpur, Gurugram, Haryana -122002
Branch Address: D-12 B 1st Floor, Chhatarpur Enclave, 100 Feet Road, New Delhi-110074
Contact: 011-43634116
Email:info@cloudeducation.co.in
Website: www.cloudeducation.co.in

3-Day Training Program: Robotics and Arduino

Total Duration: 3 Days
Daily Duration: 5 Hours

- **Session 1:** 2.5 Hours (Before Lunch)
- **Session 2:** 2.5 Hours (After Lunch)

Target Audience: University Students
Platform Used: Arduino-based Robotics Development

Day 1: Electronics Fundamentals and Arduino Introduction

Objective: Understand basic electronics, circuit building, and introduction to Arduino.

Session 1 (2.5 Hours) – Basic Electronics and Circuit Building

1. Introduction to Robotics

- What is Robotics
- Applications of Robotics in industry and research
- Overview of robotics components (controller, sensors, actuators)

2. Basic Electrical Concepts

- Voltage
- Current
- Resistance
- Power distribution in electronic circuits

3. Breadboard Fundamentals

- Structure of a breadboard
- Power rails and terminal strips
- Building circuits without soldering

4. Electronic Components

- Resistors
- LEDs
- Potentiometers
- Switches
- Push buttons

Practical Activities

Students build simple circuits:

- Power distribution on breadboard
- LED glowing circuit
- LED brightness control using potentiometer
- LED control using switch and push button

Session 2 (2.5 Hours) – Introduction to Arduino

5. Arduino Platform Overview

- What is Arduino
- Types of Arduino boards
- Arduino architecture and pin configuration

6. Arduino Development Environment

- Installing Arduino IDE
- Understanding Arduino interface
- Uploading first program

7. Arduino Programming Basics

- Structure of Arduino program
  - setup() function
  - loop() function
- Digital input and output
- Controlling LED using Arduino

Practical Activities

Students perform:

- LED blinking using Arduino
- LED control using push button
- Reading analog value from potentiometer

Day 2: Sensors and Arduino Programming

Objective: Learn about sensors, interfacing with Arduino, and writing code for sensor data.

Session 1 (2.5 Hours) – Sensor Fundamentals

1. Introduction to Sensors

- What are sensors
- Types of sensors
  - Analog sensors
  - Digital sensors
- Role of sensors in robotics and automation

2. Common Sensors Used in Robotics

Introduction to commonly used sensors such as:

- Light sensors
- Temperature sensors
- Motion sensors
- Distance sensors

3. Sensor Interfacing

- Connecting sensors to Arduino
- Analog vs digital input
- Reading sensor values

Practical Activities

Students perform:

- Connecting sensors to Arduino
- Reading sensor values using Arduino code
- Displaying sensor data in serial monitor

Session 2 (2.5 Hours) – Sensor Programming and Integration

4. Writing Programs for Sensors

- Reading analog data
- Reading digital signals
- Using conditional statements with sensors

5. Sensor Integration

- Combining sensors with actuators
- Basic automation logic

Practical Activities

Students implement:

- Sensor-based LED control
- Motion-based control system
- Distance measurement using ultrasonic sensor

Day 3: Robotics Project Development

Objective: Apply learned concepts to build robotics-based systems.

Session 1 (2.5 Hours) – Actuators and Control Systems

1. Introduction to Actuators

- What are actuators
- Types of actuators used in robotics
- Role of relays in controlling high-power devices

2. Relay Module Interfacing

- Understanding relay operation
- Connecting relay with Arduino
- Controlling external devices

Practical Activities

Students perform:

- Relay-based device control
- Automation using sensor inputs

Session 2 (2.5 Hours) – Project Development

3. Project Design and Implementation

Students design and implement a robotics-based system using:

- Arduino
- Sensors
- Relay modules
- Ultrasonic sensors

Students will:

- Design circuit on breadboard
- Write Arduino program
- Integrate sensors and actuators
- Test and troubleshoot the system

Project selection and implementation can be **based on student preference**, allowing flexibility in designing different automation or robotics systems.

Training Outcome

After completing the training students will be able to:

- Understand basic electronics and circuit design
- Build circuits using breadboard and electronic components
- Program Arduino boards
- Interface sensors with microcontrollers
- Read and process sensor data
- Control actuators such as LEDs and relays
- Design and implement basic robotics systems